

VAE

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Abstract

In this practice, I implement VAE model[1] to generate MNIST dataset and adjust the hyper parameters to optimize the reconstruction loss of VAE. Then I analysis the distribution of latent vector in case of $z_dim=1$ or 2 by sampling. Finally I got my best VAE model.

1 Introduction

Variational AutoEncoders (VAE) is a method for generative modeling, whose goal is to build the bi-bridge between normal distribution and data distribution.

1.1 Priciple

The pipeline of VAE is shown in Figure 1

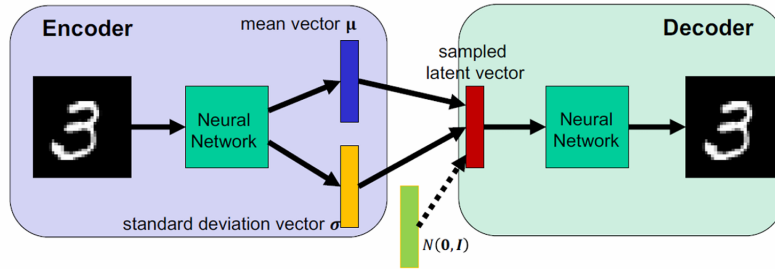


Figure 1: Pipeline

1. Encoders[2] develop a compact representation $E(x)$ of the input data(28 * 28 binary images of MNIST) at the hidden as vector z . And the hidden vector z is expected to belong to a Gaussian distribution $N(\mu, \sigma^2)$

As we enforce the distribution of the latent variables to be close to the prior distribution, typically a standard normal distribution. This regularization helps in learning a well-formed latent space and ensures that the latent variables do not deviate significantly from a prior distribution, which is often chosen to be $\mathcal{N}(0, I)$ for simplicity.

The KL divergence between two normal distributions is calculated as follows:

$$KLD(p, q) = -\log \sigma^2 + \frac{\sigma^2 + \mu^2}{2} - \frac{1}{2}$$

where μ and σ^2 typically represent the parameters of the latent variables' distribution. And it is known as the *KL Loss*

2. Decoder attempts to map the latent variables(i.e. z) from the noisy and compressed encoding back to the data space. The goal is to minimize the difference between the input X and its reconstruction $D(E(x))$, formally given by:

$$\min_{E,D} f(X, E, D) = \|X - D(E(X))\|^2$$

This difference is known as the *Reconstruction Loss*.

And the training loss of VAE is defined as a weighted combination of reconstruction loss and KL loss:

$$loss = recon_loss + \beta KL_loss$$

3. A key innovation of VAE is the reparameterization trick, which allows gradients to be back-propagated through stochastic nodes. Instead of sampling z directly from $P(z|X)$, we sample ϵ from a standard normal distribution and compute:

$$z = \mu + \sigma \odot \epsilon$$

This transformation maintains the stochastic nature of z while making the sampling differentiable, a crucial step for training VAE using gradient-based optimization methods.

1.2 model

I implement my VAE model as Figure 2

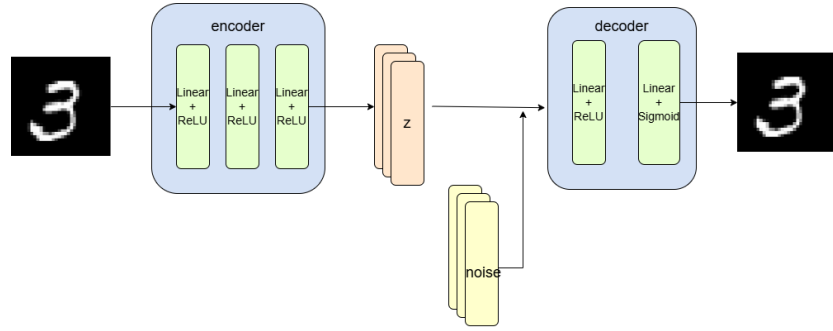


Figure 2: model

In my baseline model, the encoder is implemented by an 3-layer fully connected layer with ReLU activation, which can map 784d image to 50d hidden vector. And we use two Linear layers to produce μ and Σ respectively. The decoder is a two-layer-MLP followed by a Sigmoid activation that can recover a 784d image from a latent point. For optimizer,

2 Experiment

2.1 Baseline performance

I use Adam optimizer with learning rate as 1e-5. The loss consists of reconstruction loss which is BCELoss, and regularization loss which is the KL-distance between $N(\mu, \sigma)$ and $N(0, I)$.

My basic model reconstruct the Figure 3

2.2 Reconstruction loss optimization

I choose latent dimension, encoder layers, hidden size, and *beta* in loss function to get my best model (with less reconstruction loss). The details are shown in Table 1

1. z dim: As the dimension of latent variable represents the capability of model, I change it from 20 to 200, however, higher z dimension didn't show less reconstruction loss.
2. Number of encoder layers: as the number of encoder layer increases, reconstruction loss increases and KL loss decreases, and 3 layer got a balance between them to have a better total loss.

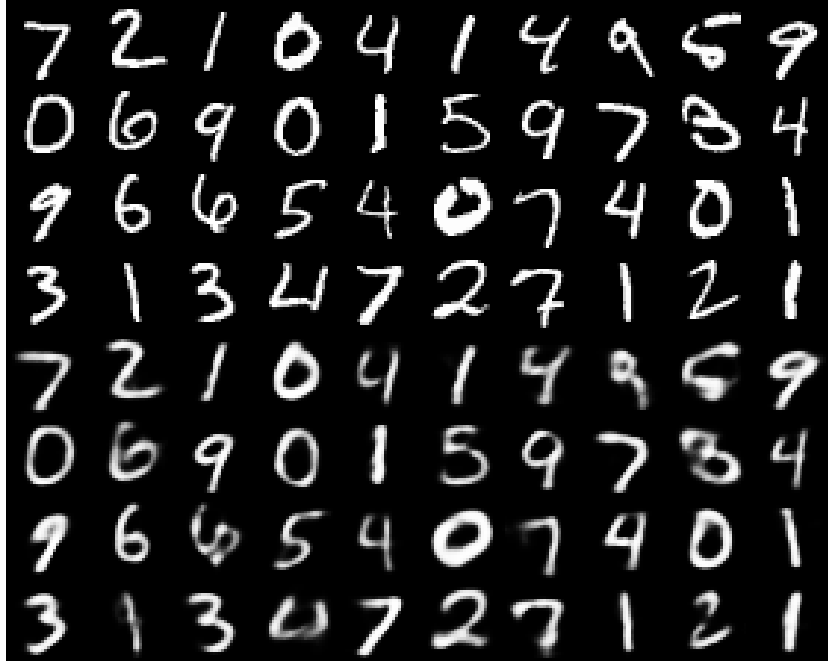


Figure 3: Baseline performance

3. Hidden size determines the intermediate vector in Encoder and Decoder. Larger hidden size results in lower reconstruction loss, but higher training KL-loss.
4. Beta: The total loss of VAE consists of two parts: reconstruction loss and regularization loss (KL loss), I adjust parameter to weight these two items. As beta increases, the reconstruction loss is less significant and increasing, while the KL loss is more important and decreasing.

2.3 z dim =1 analysis

I visualize the result when dimension is less than 3 to better my understanding of the latent variables.

Figure 4a shows the reconstruction result as z varieties from -5 to 5.

- When z is closer to -5, the reconstruction is more likely to be 1/7.
- When z is closer to 5, the reconstruction is more likely to be 0
- When z is closer to 0, the reconstruction is blurring to be 8/9/3/6

The reconstruction result is compared with ground truth in Figure 4b

The relationship between the value of z and the label of data is visualized in Figure 4c. There is apparently continuity of the same label.

2.4 z=2 analysis

Figure 5a shows the reconstruction result as z varieties from -5 to 5 in both x and y scalars. It can be roughly seen that each number has its own probability space interval.

The reconstruction result is compared with ground truth in Figure 5b

The relationship between the value of z and the label of data is visualized in Figure 5c

3 Conclusion

When number of encoder layer = 3, hidden dim =2000, beta=0.1, z - dim =200, I got my best VAE model after 200 training epochs. Its performance is shown in Figure 6 with loss=2352.53, reconstruction loss=2013.17, KL loss=3393.62.

model	loss	recon loss	kl loss
baseline	4031.10	3086.54	944.56
en-layer 2	4615.56	3351.48	632.03
en-layer 3*	4031.10	3086.54	944.56
en-layer 5	4711.29	3697.20	507.04
hid-dim 200	5083.10	4146.87	468.11
hid-dim 1000	5020.86	3796.67	612.09
hid-dim 2000	4740.61	3372.27	684.17
beta 0.1	2844.25	2504.71	3392.62
beta 0.5	3442.55	2722.93	1439.23
beta 1*	4031.10	3086.54	944.56
beta 2	4718.94	3518.19	600.37
z dim 20	4027.33	3067.01	960.32
z dim 50*	4031.10	3086.54	944.56
z dim 100	4435.2	3409.28	1025.92
z dim 200	4075.66	3111.81	963.84

Table 1: Optimization

* means the model keep the baseline setting, en-layer means the number of layers in encoder, hid-dim means the hidden dimension of decoder, beta is the hyper parameter in loss computation, and z-dim is the dimension of latent vector z.

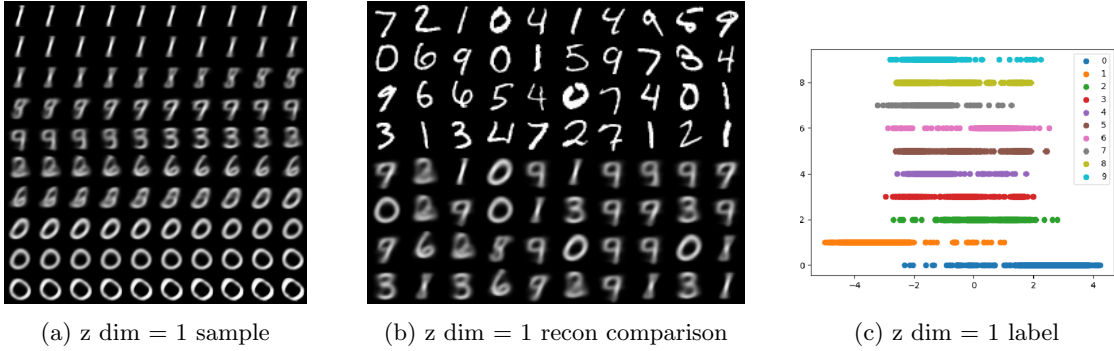


Figure 4: Visualization of different aspects of z dim=1

References

- [1] Carl Doersch. Tutorial on variational autoencoders. *arXiv preprint arXiv:1606.05908*, 2016.
- [2] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.

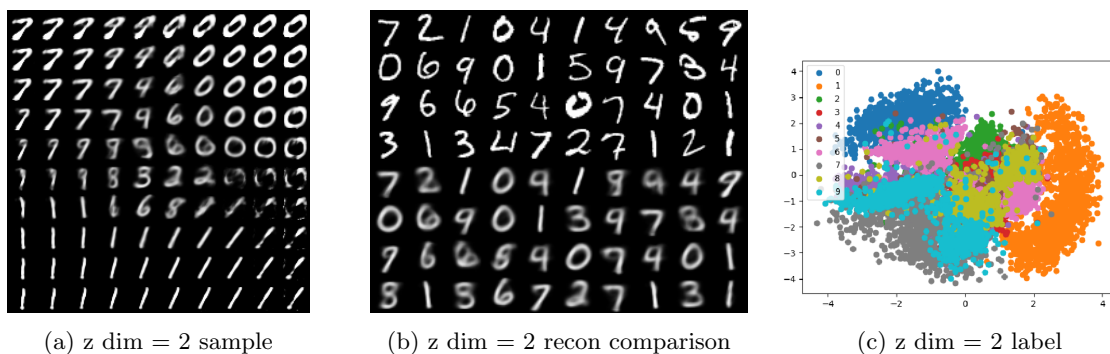


Figure 5: Visualization of different aspects of z dim=2



Figure 6: Best